

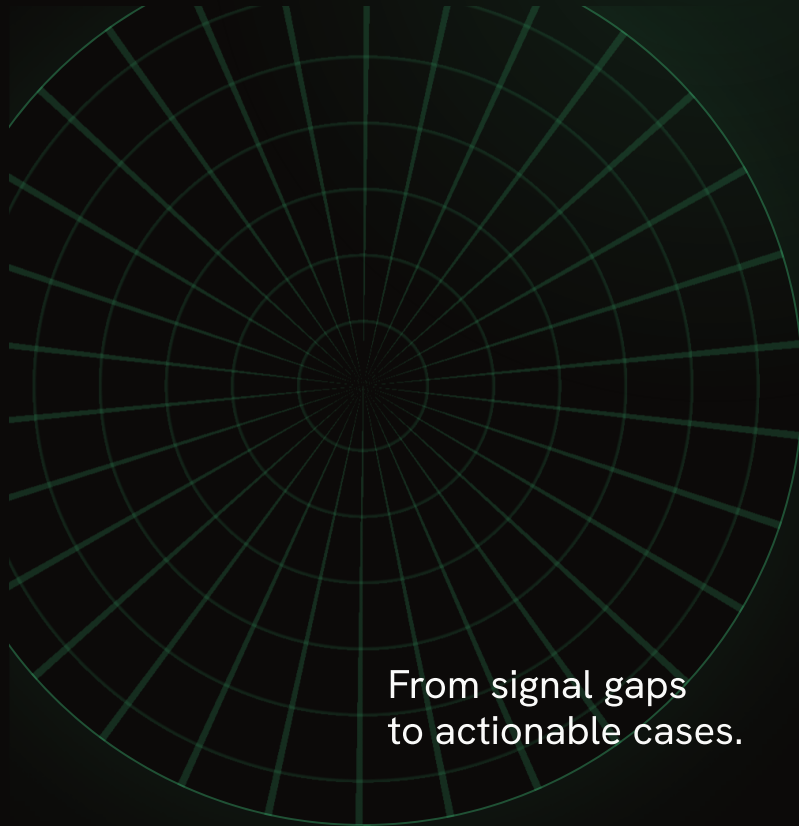


Wake

AI

Making dark shipping
visible.

SEPT 2026



When a vessel goes dark, the trail goes cold.



Automatic Identification System (AIS) tracks a vessel's location, speed, and bearing to avoid collisions.

However, AIS can be faulty in poor conditions or when a vessel disables it.

These AIS blackouts can hide ship-to-ship transfers, illegal fishing, and sanctions evasion.

A massive Blind Spot

Left in the wake.

43%

est. global seaborne crude exports,
'17-'23

558

avg. suspected dark tankers/year,
~25% of global fleet

7.8M

metric tons crude/month,
'17-'23

~\$334B

est. crude value/year,
avg. '17-'23 Brent

This 'dark shipping' drives deflationary growth, allowing adversaries to leverage discounted oil and boost industrial output.

Sources: [Fernandez-Villaverde, Li, Xu & Zanetti \(2025\)](#); [U.S. EIA Brent history](#).

An ensemble of factors.

01

AIS gaps

When and where a vessel disappears

02

Geography

Risky routes and sanctioned-port proximity

03

Behavior

Speed, heading and voyage anomalies

04

Proximity

Potential ship-to-ship transfers

AIS gaps alone are not sufficient. Composite factors provide differentiated reporting.

Multi-method detection.

01 / BEHAVIORAL BASELINE

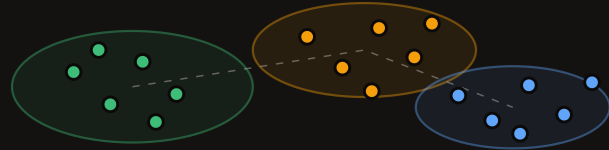
Detect deviations from a vessel's own pattern.



Use a Taylor approximation of known rendezvous behavior to estimate likely dark-rendezvous paths.

02 / PROXIMITY CLUSTERING

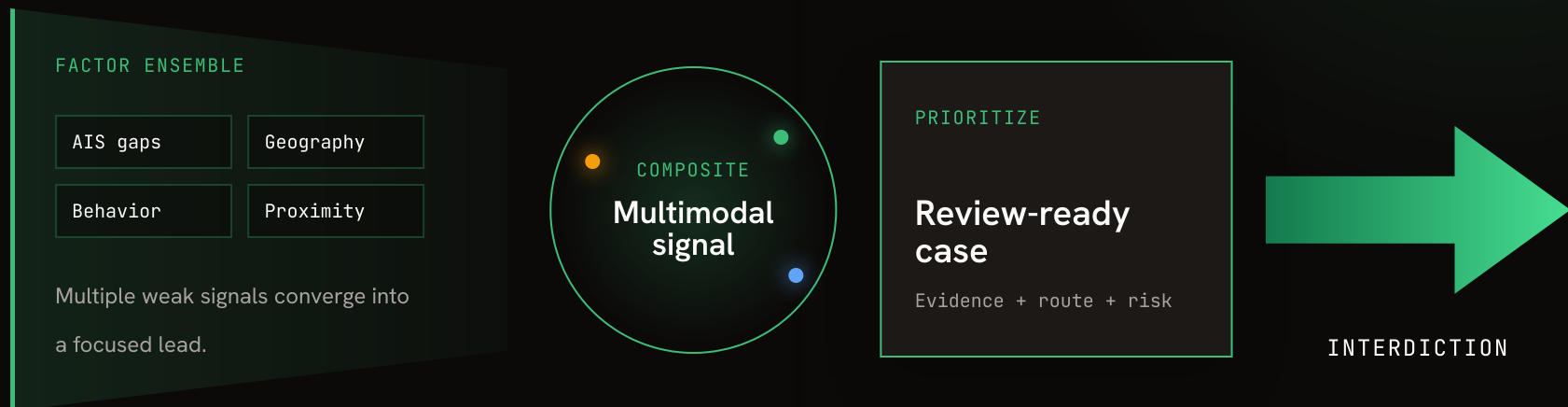
Surface unusual vessel-to-vessel proximity.



Cluster nearby vessels to reveal repeated, close-range interactions that merit review for coordinated activity.

BEHAVIOR + PROXIMITY = COMPOSITE RISK SIGNAL

From signal to review-ready lead.



Our system enables analysts, maritime-security teams, and enforcement partners to proactively identify high-risk activity and strengthen maritime domain awareness before it becomes an incident.

• WAKE AI

When ships go dark intelligence should not.

Making dark shipping visible.
